Question 1

Que-1:
Create 2 Public Docker Hub registries named
cloudethix_master_nginx_yourname & cloudethix_release_nginx_yourname.

- Clone below repository on your system.
  https://github.com/zembutsu/docker-sample-nginx.git
- Initialize a local repository & copy the code from above repo to your local repository in master branch and then create below branches.
  release
  main
  hotfix
- Once code is copied to local repository,from master branch update the index.html and add word "Cloudethix Master Branch Nginx" and build the docker image & add meaningful tags and push to Docker Hub registry cloudethix_master_nginx_yourname.
- Also from release branch update the index.html and add word "Cloudethix Release Branch Nginx" and build the docker image & add meaningful tags and push to Docker Hub registry cloudethix_release_nginx_yourname.
- Once Images are copied to Docker hub registries, switch to the main branch.
- In main branch create directory named kube/clusterIP & inside kube directory create file named master_pod.yaml with pod name master_nginx & with label master_nginx & add image that you have pushed in Docker Hub registry cloudethix_master_nginx_yourname.
- Also create a file release_pod.yaml with pod name release_nginx & with label release_nginx & add image that you have pushed in Docker Hub registry cloudethix_release_nginx_yourname.
- Create a file called cluster_ip-service.yaml with service name cloudethix_clusterip and with Type clusterIP.
- Then, select the pod with label release_nginx in service
  Create all these three resources in your k8s cluster.
- Now, access master_nginx pod shell & curl the master_nginx pod & check the result.
- Also try to curl release_nginx pod with DNS name & check the result.
- Then curl the clusterip service with its name and check the result.
- Finally, create a GITHUB remote repository named cloudethix-k8s-yourname and push all the branches to the remote repository.
- Take all screenshots and create a well formatted document.

System Output

```
payal_kharat@LAPTOP-# git clone https://github.com/zembutsu/docker-sample-
nginx.git
Cloning into 'docker-sample-nginx'...
remote: Enumerating objects: 22, done.
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 22 (delta 7), reused 6 (delta 6), pack-reused 10 (from 1)
```

```
Receiving objects: 100% (22/22), done.
Resolving deltas: 100% (7/7), done.
payal_kharat@LAPTOP-# git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in
/mnt/d/CloudEthix_Internship/Kubernetes/Assignments/Assignment-2/Que-1/.git/
-------------------------------------------------------------------------
payal_kharat@LAPTOP-# cp -rf docker-sample-nginx/* .
payal_kharat@LAPTOP-# ls
Dockerfile  LICENSE  README.md  default.conf  docker-sample-nginx  index.html
payal_kharat@LAPTOP-# git status
On branch master
No commits yet
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Dockerfile
        LICENSE
        README.md
        default.conf
        docker-sample-nginx/
        index.html
nothing added to commit but untracked files present (use "git add" to track)
----------------------------------------------------------------------------
payal_kharat@LAPTOP-# git add .
warning: adding embedded git repository: docker-sample-nginx
hint: You've added another git repository inside your current repository.
hint: Clones of the outer repository will not contain the contents of
hint: the embedded repository and will not know how to obtain it.
hint: If you meant to add a submodule, use:
hint:
hint:   git submodule add <url> docker-sample-nginx
hint:
hint: If you added this path by mistake, you can remove it from the
hint: index with:
hint:
hint:   git rm --cached docker-sample-nginx
hint:
hint: See "git help submodule" for more information.
hint: Disable this message with "git config set advice.addEmbeddedRepo false"
payal_kharat@LAPTOP-# git commit -m " nginx-sample application"
[master (root-commit) 7df6705]  nginx-sample application
 6 files changed, 47 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 LICENSE
 create mode 100644 README.md
```

```
 create mode 100644 default.conf
 create mode 160000 docker-sample-nginx
 create mode 100644 index.html
----------------------------------------------------------------------------
payal_kharat@LAPTOP-# git branch main
payal_kharat@LAPTOP-# git branch release
payal_kharat@LAPTOP-# git branch hotfix
payal_kharat@LAPTOP-# git branch
  hotfix
  main
* master
  release
----------------------------------------------------------------------------
payal_kharat@LAPTOP-# podman build -t cloudethix-master-nginx .
STEP 1/3: FROM nginx:alpine
STEP 2/3: COPY default.conf /etc/nginx/conf.d/
--> Using cache
bfba633494f51cc913622ffd1987833f35e1c8cff0ccf02885633eeca04dc5a0
--> bfba633494f5
STEP 3/3: COPY index.html /usr/share/nginx/html/
COMMIT cloudethix-master-nginx
--> caf1290ae9d7
Successfully tagged localhost/cloudethix-master-nginx:latest
caf1290ae9d7044437a1f170ab6b175cd1651fd93567fa10f5ef3bc2250d0b48
payal_kharat@LAPTOP-# podman login docker.io
Username: payalkharat
Password:
Login Succeeded!
------------------------------------------------------------------------------
payal_kharat@LAPTOP-# podman tag localhost/cloudethix-master-nginx
docker.io/payalkharat/cloude
thix_master_nginx_payal:master-v1
payal_kharat@LAPTOP-# podman push
docker.io/payalkharat/cloudethix_master_nginx_payal:master-v1
Getting image source signatures
Copying blob 0d62d88506ba skipped: already exists
Copying blob 1ed8e39a7434 skipped: already exists
Copying blob d94291c26261 skipped: already exists
Copying blob 8cdfe4f23778 skipped: already exists
Copying blob 55afa1ecc21d skipped: already exists
Copying blob 0ea935727878 skipped: already exists
Copying blob da4dea1b00af skipped: already exists
Copying blob e011cd894381 done   |
Copying blob 31ecba1cce22 done   |
Copying blob fb597529c916 skipped: already exists
Copying config caf1290ae9 done   |
Writing manifest to image destination
----------------------------------------------------------------------------
payal_kharat@LAPTOP-# git checkout release
Switched to branch 'release'
payal_kharat@LAPTOP-# vim index.html
payal_kharat@LAPTOP-# git add index.html
payal_kharat@LAPTOP-# git commit -m "Update nginx for Cloudethix release
branch"
[release c1fd64f] Update nginx for Cloudethix release branch
 1 file changed, 1 insertion(+)
payal_kharat@LAPTOP-# podman build -t cloudethix-release-nginx .
STEP 1/3: FROM nginx:alpine
STEP 2/3: COPY default.conf /etc/nginx/conf.d/
--> Using cache
```

```
bfba633494f51cc913622ffd1987833f35e1c8cff0ccf02885633eeca04dc5a0
--> bfba633494f5
STEP 3/3: COPY index.html /usr/share/nginx/html/
COMMIT cloudethix-release-nginx
--> 611d02a8945b
Successfully tagged localhost/cloudethix-release-nginx:latest
611d02a8945b16a022f539e7a62688a4c0336161788c352834df9a57ca413e4a
-----------------------------------------------------------------------------
payal_kharat@LAPTOP-# podman tag localhost/cloudethix-release-nginx
docker.io/payalkharat/cloud
ethix_release_nginx_payal:release-v1
payal_kharat@LAPTOP-# podman push
docker.io/payalkharat/cloudethix_release_nginx_payal:release-v1
Getting image source signatures
Copying blob 1ed8e39a7434 skipped: already exists
Copying blob d94291c26261 skipped: already exists
Copying blob 0d62d88506ba skipped: already exists
Copying blob 8cdfe4f23778 skipped: already exists
Copying blob 55afa1ecc21d skipped: already exists
Copying blob 0ea935727878 skipped: already exists
Copying blob e198772e16f4 done   |
Copying blob da4dea1b00af skipped: already exists
Copying blob fb597529c916 skipped: already exists
Copying blob b83fc70afc5e skipped: already exists
Copying config 611d02a894 done   |
Writing manifest to image destination
-----------------------------------------------------------------------------
payal_kharat@LAPTOP-# git branch
  hotfix
* main
  master
  release
payal_kharat@LAPTOP-# mkdir -p kube/clusterIP
payal_kharat@LAPTOP-# cd kube/
payal_kharat@LAPTOP-# cd clusterIP/
-----------------------------------------------------------------------------
payal_kharat@LAPTOP-# cat master-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: master-nginx-revision1
  namespace: cloudethix-1
  labels:
    app: master-nginx-revision1
spec:
  containers:
    - name: master-nginx-revision1
      image: docker.io/payalkharat/cloudethix_master_nginx_payal:master-v1
      ports:
        - containerPort: 80
------------------------------------------------------------------------------
payal_kharat@LAPTOP-# kubectl apply -f master-pod.yaml
pod/master-nginx-revision1 created
payal_kharat@LAPTOP-# kubectl get pods -n cloudethix-1
NAME                     READY   STATUS    RESTARTS   AGE
master-nginx-revision1   1/1     Running   0          20s
------------------------------------------------------------------------------
payal_kharat@LAPTOP-# cat release-pod.yaml
apiVersion: v1
kind: Pod
```

```
metadata:
  name: release-nginx-revision1
  namespace: cloudethix-1
  labels:
    app: release-nginx-revision1
spec:
  containers:
    - name: release-nginx-revision1
      image: docker.io/payalkharat/cloudethix_release_nginx_payal:release-v1
      ports:
        - containerPort: 80
------------------------------------------------------------------------------
payal_kharat@LAPTOP-# kubectl apply -f release-pod.yaml
pod/release-nginx-revision1 created
payal_kharat@LAPTOP-# kubectl get pods -n cloudethix-1
NAME                       READY   STATUS    RESTARTS   AGE
master-nginx-revision1     1/1     Running   0          20h
release-nginx-revision1    1/1     Running   0          20h
------------------------------------------------------------------------------
payal_kharat@LAPTOP-# kubectl apply -f cluster-ip-service.yaml
service/cloudethix-clusterip-revision1 created
payal_kharat@LAPTOP-# kubectl get svc -n cloudethix-1
NAME                             TYPE        CLUSTER-IP     EXTERNAL-IP
PORT(S)   AGE
cloudethix-clusterip-revision1   ClusterIP   10.99.78.169   <none>
80/TCP    20h
------------------------------------------------------------------------------
payal_kharat@LAPTOP-# kubectl exec -it master-nginx-revision1 -n cloudethix-1
-- /bin/sh
/ # curl localhost
<html>
<body>
        <h1>Host: master-nginx-revision1</h1>
        Version: 1.1
        <h1> Cloudethix Master Branch Nginx </h1>
</body>
</html>
/ # curl http://cloudethix-clusterip-revision1
<html>
<body>
        <h1>Host: release-nginx-revision1</h1>
        Version: 1.1
        <h1>Cloudethix Release Branch Nginx</h1>
</body>
</html>
```

Question 2

Question-2:
In the main branch of your local repository create a directory kube/NodePort.

- Create below files from below url. Please make sure you will create NodePort service with port 30008 instead of loadbalancer.
  https://kubernetes.io/docs/tasks/access-application-cluster/connecting-frontend-backend/.
  backend-deployment.yaml
  backend-service.yaml

frontend-deployment.yaml
frontend-NodePort-service.yaml

- Once files are created , create all the resources in your k8s cluster.
- Access all public ips with port 30008 in the browser and then check the result.
- Finally, push all the latest code to the remote repository.
- Take all screenshots and create a well formatted document

Question 2 — Execution Output

```
payal_kharat@LAPTOP-# git branch
  hotfix
* main
  master
  release
payal_kharat@LAPTOP-# cd kube/
payal_kharat@LAPTOP-# mkdir NodePort
payal_kharat@LAPTOP-# cd NodePort/
------------------------------------------------------------------------------
payal_kharat@LAPTOP-# cat backend-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: backend-revision1
  namespace: cloudethix-1
spec:
  selector:
    matchLabels:
      app: hello
      tier: backend
      track: stable
  replicas: 3
  template:
    metadata:
      labels:
        app: hello
        tier: backend
        track: stable
    spec:
      containers:
        - name: hello
          image: "gcr.io/google-samples/hello-go-gke:1.0"
          ports:
            - name: http
              containerPort: 80
------------------------------------------------------------------------------
payal_kharat@LAPTOP-# cat backend-service.yaml
apiVersion: v1
kind: Service
metadata:
  name: hello
  namespace: cloudethix-1
spec:
  selector:
    app: hello
    tier: backend
  ports:
  - protocol: TCP
```

```
    port: 80
    targetPort: http
------------------------------------------------------------------------------
payal_kharat@LAPTOP-# cat frontend-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: frontend-revision1
  namespace: cloudethix-1
spec:
  selector:
    matchLabels:
      app: hello
      tier: frontend
      track: stable
  replicas: 1
  template:
    metadata:
      labels:
        app: hello
        tier: frontend
        track: stable
    spec:
      containers:
        - name: nginx
          image: "gcr.io/google-samples/hello-frontend:1.0"
          lifecycle:
            preStop:
              exec:
                command: ["/usr/sbin/nginx","-s","quit"]
----------------------------------------------------------------------------
payal_kharat@LAPTOP-# cat frontend-service.yaml
apiVersion: v1
kind: Service
metadata:
  name: frontend
  namespace: cloudethix-1
spec:
  selector:
    app: hello
    tier: frontend
  type: NodePort
  ports:
  - protocol: "TCP"
    port: 80
    targetPort: 80
    nodePort: 30008
-----------------------------------------------------------------------------
payal_kharat@LAPTOP-# kubectl apply -f backend-deployment.yaml
deployment.apps/backend-revision1 created
payal_kharat@LAPTOP-# kubectl apply -f frontend-deployment.yaml
deployment.apps/frontend-revision1 created
payal_kharat@LAPTOP-# kubectl get deployment
NAME                 READY   UP-TO-DATE   AVAILABLE   AGE
backend-revision1    3/3     3            3           65s
frontend-revision1   1/1     1            1           53s
------------------------------------------------------------------------------
payal_kharat@LAPTOP-# kubectl get pods
NAME                                  READY   STATUS    RESTARTS   AGE
backend-revision1-7cdc9fcc87-f5s8p    1/1     Running   0          80s
```

```
backend-revision1-7cdc9fcc87-p8q6v    1/1     Running   0          80s
backend-revision1-7cdc9fcc87-vfg8g    1/1     Running   0          80s
frontend-revision1-7db75b888d-2cxjk   1/1     Running   0          68s
------------------------------------------------------------------------------
payal_kharat@LAPTOP-# kubectl apply -f frontend-service.yaml
service/frontend created
payal_kharat@LAPTOP-# kubectl apply -f backend-service.yaml
service/hello created
payal_kharat@LAPTOP-# kubectl get svc
NAME       TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
frontend   NodePort    10.107.150.184   <none>        80:30008/TCP   49s
hello      ClusterIP   10.104.60.127    <none>        80/TCP         39s
------------------------------------------------------------------------------
payal_kharat@LAPTOP-# curl http://localhost:8080
{"message":"Hello"}
payal_kharat@LAPTOP-# git status
On branch main
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        ../
nothing added to commit but untracked files present (use "git add" to track)
payal_kharat@LAPTOP-V1NV0QFT:NodePort$ ls
backend-deployment.yaml  backend-service.yaml  frontend-deployment.yaml
frontend-service.yaml
------------------------------------------------------------------------------
payal_kharat@LAPTOP-# git add .
payal_kharat@LAPTOP-#git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   backend-deployment.yaml
        new file:   backend-service.yaml
        new file:   frontend-deployment.yaml
        new file:   frontend-service.yaml
payal_kharat@LAPTOP-# git commit -m " Frontend and backend files are added"
[main b5a95b2]  Frontend and backend files are added
 4 files changed, 79 insertions(+)
 create mode 100644 kube/NodePort/backend-deployment.yaml
 create mode 100644 kube/NodePort/backend-service.yaml
 create mode 100644 kube/NodePort/frontend-deployment.yaml
 create mode 100644 kube/NodePort/frontend-service.yaml
------------------------------------------------------------------------------
payal_kharat@LAPTOP-# git remote add origin
https://github.com/CloudEthix987/Cloudethix-Nginx-Payal.git
payal_kharat@LAPTOP-V1NV0QFT:NodePort$ git push -u origin main
Username for 'https://github.com': CloudEthix987
Password for 'https://CloudEthix987@github.com':
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 12 threads
Compressing objects: 100% (14/14), done.
Writing objects: 100% (15/15), 2.50 KiB | 75.00 KiB/s, done.
Total 15 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
remote:
remote: Create a pull request for 'main' on GitHub by visiting:
remote:      https://github.com/CloudEthix987/Cloudethix-Nginx-
Payal/pull/new/main
remote:
To https://github.com/CloudEthix987/Cloudethix-Nginx-Payal.git
```

```
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Question 3

Question-3:
Create any 2 pods and assign them to different worker nodes with
nodeName property.

Execution Output

```
payal_kharat@LAPTOP-# kubectl get nodes
NAME                      STATUS   ROLES           AGE    VERSION
mycluster-control-plane   Ready    control-plane   146m   v1.34.0
mycluster-worker          Ready    <none>          145m   v1.34.0
mycluster-worker2         Ready    <none>          145m   v1.34.0
----------------------------------------------------------------------------
payal_kharat@LAPTOP-# cat pod1.yaml
apiVersion: v1
kind: Pod
metadata:
  name: pod1
  namespace: cloudethix-1
spec:
  nodeName: mycluster-worker
  containers:
    - name: nginx
      image: nginx:alpine
      ports:
        - containerPort: 80
----------------------------------------------------------------------------
payal_kharat@LAPTOP-# cat pod2.yaml
apiVersion: v1
kind: Pod
metadata:
  name: pod2
  namespace: cloudethix-1
spec:
  nodeName: mycluster-worker2
  containers:
    - name: nginx
      image: nginx:alpine
      ports:
        - containerPort: 80
----------------------------------------------------------------------------
payal_kharat@LAPTOP-# kubectl apply -f pod1.yaml
pod/pod1 created
payal_kharat@LAPTOP-# kubectl apply -f pod2.yaml
pod/pod2 created
payal_kharat@LAPTOP-# kubectl get pod -o wide
NAME   READY   STATUS    RESTARTS   AGE   IP            NODE
NOMINATED NODE   READINESS GATES
pod1   1/1     Running   0          12s   10.244.1.7    mycluster-worker
<none>           <none>
pod2   1/1     Running   0          7s    10.244.2.10   mycluster-worker2
<none>           <none>
```

Question 4

Question-4:
Label both worker nodes such as worker-0 node as cloudethix-k8s-00 &
worker-1 node as cloudethix-k8s-01.

- Once nodes are labeled, create pod00.yaml file and schedule the pod on worker-0 node with nodeSelector property. Also create one more file named pod01.yaml & schedule the pod on worker-1 node.

Execution Output

```
payal_kharat@LAPTOP-# kubectl get nodes
NAME                        STATUS   ROLES           AGE    VERSION
mycluster-control-plane     Ready    control-plane   158m   v1.34.0
mycluster-worker            Ready    <none>          157m   v1.34.0
mycluster-worker2           Ready    <none>          157m   v1.34.0
----------------------------------------------------------------------------
payal_kharat@LAPTOP-# kubectl label nodes mycluster-worker cloudethix-k8s-
00=worker-0
node/mycluster-worker labeled
payal_kharat@LAPTOP-# kubectl label nodes mycluster-worker2 cloudethix-k8s-
01=worker-1
node/mycluster-worker2 labeled
----------------------------------------------------------------------------
payal_kharat@LAPTOP-# kubectl get nodes --show-labels
NAME                        STATUS   ROLES           AGE    VERSION   LABELS
mycluster-control-plane     Ready    control-plane   164m   v1.34.0
beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,kubernetes.io/arch=a
md64,kubernetes.io/hostname=mycluster-control-
plane,kubernetes.io/os=linux,node-role.kubernetes.io/control-
plane=,node.kubernetes.io/exclude-from-external-load-balancers=
mycluster-worker            Ready    <none>          164m   v1.34.0
beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,cloudethix-k8s-
00=worker-0,kubernetes.io/arch=amd64,kubernetes.io/hostname=mycluster-
worker,kubernetes.io/os=linux
mycluster-worker2           Ready    <none>          164m   v1.34.0
beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,cloudethix-k8s-
01=worker-1,kubernetes.io/arch=amd64,kubernetes.io/hostname=mycluster-
worker2,kubernetes.io/os=linux
----------------------------------------------------------------------------
payal_kharat@LAPTOP-# cat pod00.yaml
apiVersion: v1
kind: Pod
metadata:
  name: pod00
  namespace: cloudethix-1
spec:
  nodeSelector:
    cloudethix-k8s-00: worker-0
  containers:
    - name: nginx
      image: nginx:alpine
      ports:
        - containerPort: 80
----------------------------------------------------------------------------
payal_kharat@LAPTOP-# cat pod01.yaml
apiVersion: v1
kind: Pod
metadata:
  name: pod01
```

```
  namespace: cloudethix-1
spec:
  nodeSelector:
    cloudethix-k8s-01: worker-1
  containers:
    - name: nginx
      image: nginx:alpine
      ports:
        - containerPort: 80
----------------------------------------------------------------------------
payal_kharat@LAPTOP-# kubectl apply -f pod00.yaml
pod/pod00 created
payal_kharat@LAPTOP-# kubectl apply -f pod01.yaml
pod/pod01 created
payal_kharat@LAPTOP-# kubectl get pods -o wide
NAME    READY   STATUS    RESTARTS   AGE   IP            NODE
NOMINATED NODE   READINESS GATES
pod00   1/1     Running   0          13s   10.244.1.8    mycluster-worker
<none>           <none>
pod01   1/1     Running   0          9s    10.244.2.11   mycluster-worker2
<none>           <none>
```

Question 5

Question-5
Clone the below repo locally & create DaemonSet from directory
DaemonSet101.

Execution Output

```
payal_kharat@LAPTOP-# git clone https://github.com/collabnix/kubelabs.git
Cloning into 'kubelabs'...
remote: Enumerating objects: 14031, done.
remote: Counting objects: 100% (707/707), done.
remote: Compressing objects: 100% (147/147), done.
remote: Total 14031 (delta 637), reused 560 (delta 560), pack-reused 13324
(from 5)
Receiving objects: 100% (14031/14031), 61.81 MiB | 899.00 KiB/s, done.
Resolving deltas: 100% (4181/4181), done.
Updating files: 100% (7374/7374), done.
----------------------------------------------------------------------------
payal_kharat@LAPTOP-# ls
kubelabs
payal_kharat@LAPTOP-# cd kubelabs/
payal_kharat@LAPTOP-# ls
 201                    Gemfile                        LICENSE
Secrets101            assets                         kube101.md
 AKS101                 Gemfile.lock                   LKE101
Security101           bluegreen                      kubectl-for-docker.md
 Autoscaler101          GitLab101                      'Labels&Selectors'
ServiceCatalog101     bootstrap.sh                   namespace.yaml
 CONTRIBUTING.md        GitOps101                      Labels-and-Selectors
ServiceMesh101        cka                            nodejs.md
 Chaos101               Helm101                        Loft101
Services101           context                        pods101
 'Cheat Sheets'         IOT                            Logging101
Shipa101              detect.md                      pods101.png
 ClusterNetworking101   Ingress101                     ManagedKubernetes
SlidesReplicaSet101   dockerdesktopformac            portainer
```

```
 ConfigMaps101         InitContainers101            Monitoring101
Slides_Services101    dockervsk8s                  portainer.yaml
 Cronjobs101           JavaClient101                MultiTenant101
StatefulSets101       gke-setup.md                 python
 Cronjobs101.md        Jenkins101                   Network_Policies101
Strimzi101            golang                       replicaset101
 DaemonSet101          Jobs101                      Observability101
Terraform101          helm                         scripts
 Deployment101         Karpenter101                 Pods101_slides
Weave-UI.png          images                       secerts_configmaps101
 Deployment101_slides  Keda101                      RBAC101
_config.yml           img                          weave-pwk.md
 DevSpace101           KubeSphere                   README.md
_config_dev.yml       install                      weave-service.png
 DisasterRecovery101   Kubernetes_Architecture.md   ReplicationController101
_layouts              intermediate                 weave.jpg
 EKS101                Kubernetes_Intro_slides-1    ResourceQuota101
_site                 jekyll-theme-hacker.gemspec  weave.md
 GKE101                Kubezoo                      Rollouts101
ai                    k3s
 Gateway101            Kyverno101                   Scheduler101
api.md                k8sworkshop.png
-----------------------------------------------------------------------------
payal_kharat@LAPTOP-# cd DaemonSet101/
payal_kharat@LAPTOP-# ls
README.md  daemonset.yml
payal_kharat@LAPTOP-# kubectl apply -f daemonset.yml
daemonset.apps/prometheus-daemonset created
payal_kharat@LAPTOP-# kubectl get pods
NAME                         READY   STATUS    RESTARTS   AGE
prometheus-daemonset-vnnrs   1/1     Running   0          15s
```

Question 6

Question-6
Create a static pod with name cloudethix-static in your k8s cluster. Refer below link.
https://kubernetes.io/docs/tasks/configure-pod-container/static-pod-container/static-pod

Execution Output

```
payal_kharat@LAPTOP-# cat static-pod.yaml
apiVersion: v1
kind: Pod
metadata:
  name: cloudethix-static
  namespace: cloudethix-1
  labels:
    role: myrole
spec:
  containers:
    - name: web
      image: nginx
      ports:
        - name: web
          containerPort: 80
          protocol: TCP
-----------------------------------------------------------------------------
payal_kharat@LAPTOP-# docker cp cloudethix-static.yaml mycluster-control-
```

```
plane:/etc/kubernetes/manifests/cloudethix-static.yaml
Successfully copied 2.05kB to mycluster-control-
plane:/etc/kubernetes/manifests/cloudethix-static.yaml
----------------------------------------------------------------------------
payal_kharat@LAPTOP-# podman exec mycluster-control-plane ls -l
/etc/kubernetes/manifests/
total 20
-rwxrwxrwx 1 1000 1000  254 Sep  6 16:57 cloudethix-static.yaml
-rw------- 1 root root 2614 Sep  6 11:40 etcd.yaml
-rw------- 1 root root 3968 Sep  6 11:40 kube-apiserver.yaml
-rw------- 1 root root 3498 Sep  6 11:40 kube-controller-manager.yaml
-rw------- 1 root root 1726 Sep  6 11:40 kube-scheduler.yaml
payal_kharat@LAPTOP-V1NV0QFT:Que-6$ kubectl get pods
----------------------------------------------------------------------------
NAME                                        READY   STATUS    RESTARTS   AGE
cloudethix-static-mycluster-control-plane   1/1     Running   0          2m45s
```

Question 7

Question-7
Install Kubectx & kubens in your k8s cluster

Execution Output

```
Install Kubectx & kubens in your k8s cluster
payal_kharat@LAPTOP-# kubectx
kind-mycluster
payal_kharat@LAPTOP-V# kubectl config current-context
kind-mycluster
payal_kharat@LAPTOP-# kubens
cloudethix-payal
default
kube-node-lease
kube-public
kube-system
local-path-storage
payal_kharat@LAPTOP-# kubens kube-system
✓ Active namespace is "kube-system"
payal_kharat@LAPTOP-# kubens
cloudethix-payal
default
kube-node-lease
kube-public
kube-system
local-path-storage
payal_kharat@LAPTOP-# kubens default
✓ Active namespace is "default"
```